Drew Council

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Experience

Senior Software Engineer — Nov 2024 — Present
Dirac, Inc. — *New York, NY*

- **Developed high-performance assembly sequencing and path planning algorithms** using custom OpenCL kernels, reducing computation time by 10,000x.
- **Led the technical implementation of a full in-browser 3D CAD viewer**, bridging complex spatial backend data with interactive frontend visualization.
- **Managed application-wide database migrations**, meticulously executing data transitions and reducing post-migration data error rates by 80%.
- **Established the company's observability stack**, halving response times for production issues through comprehensive monitoring and logging systems.
- **Engineered scalable backend services** to support the platform's core automated work instructions engine.

Software Engineer — Jan 2022 — May 2024
BotBuilt Robotics — *Durham, NC*

- **Worked with others on an agile team** to develop software for robotic construction of houses.
- **Headed company DevOps**, including managing CI/CD, developer tools, and embedded software deployment.
- **Designed embedded hardware** for a ROS2 network, providing services for actuator and sensor control.
- **Integrated AWS hosting** for automated Docker image builds with custom logging and error reporting.

Software Subteam Lead — Jun 2020 — Jun 2022
Duke University Robotics Club — *Durham, NC*

- **Competed in annual RoboSub robot competition**, where we designed a fully autonomous submarine robot.
- **Coordinated a 25+ member agile environment team** using ROS, Docker, and Git to manage a shared codebase.
- **Implemented PID, Sensor Fusion, Computer Vision, and SMACH** to improve robot accuracy and capability.
- Earned 1st in Propulsion System, 3rd in Sensor optimization in 2021; 1st in technical report in 2021 and 2022.

Projects

Duke General Robotics Lab, Cell Robots Research — Feb 2023 — Present

- Designed omnidirectional grounded robot swarm for research in distributed control and communication.
- Researched algorithms to control a swarm of robots to perform tasks with low communication and orchestration.

Duke Product Design, "Aelevate" Bike Trainer — Sep 2022 — Dec 2022

- Used PlatformIO to write C++ embedded software to read sensors and control motors on a tilting bike trainer.
- Created a serial interface to communicate with a custom GUI application for touchscreen user control.

Duke Computer Engineering, FPGA Typeracer-style Arcade Game — Sep 2022 — Dec 2022

- Used Verilog to implement a MIPS-like pipelined processor to process serial input and generate VGA graphics.
- Implemented a head-to-head typing arcade game including display and keyboard drivers on an FPGA.

Duke Engineering First Year Design, Room Availability Detection System — Sep 2020 — May 2021

- Used a Raspberry Pi and Python to analyze and error correct PIR sensor data and detect a person's presence.
- Communicated this sensor data over a custom web API to publish room status and use statistics to a website.

Education

Duke University — Durham, NC
Bachelor of Computer Science, Electrical & Computer Engineering — *Sep 2020 — May 2024*

- Completed classes in Data Structures/Algorithms, Computer Architecture, Networking, and Robotics.

Skills

Go, Python, Rust, C, C++, TypeScript, OpenCL, SQL, DuckDB, Linux, NixOS, Docker, Git, Bash, Arduino, Verilog, Nushell, AWS